# Competing Customs Unions and Market Integration

Grain prices in the German states, 1818–1833

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# Question

# Three customs unions formed at once in one economic space

- 1828: the Prussian–Hessian union (PHC), the South German union (SGCU) and the Central German Commercial Union (CGCU) divide the German states
- Five years later most of them merge into the Zollverein (1834), the first customs union between sovereign states (Ploeckl, 2015, pp. 277–278)
- A natural experiment in **simultaneous, competing regional trade agreements**
- Germany "began as a customs union of smaller states" (Freund and Ornelas, 2010, p. 159)

# The question

What happens to market integration when several regional trade agreements form at the same time and partition an existing economic space into competing blocs?

- Inside a bloc: do markets integrate?
- At the edge between blocs: do they separate?
- How large are any differences, against aggregate shocks and distance?

# What the literature says: agreements, borders and geography

## Trade agreements and borders

- Creation and diversion (Viner, 1950): preferential liberalisation closes the price gap with partners, widens it with outsiders (Baldwin and Wyplosz, 2015, ch. 5)
- Agreements form between near neighbours that already trade (Freund and Ornelas, 2010, p. 144); formation is endogenous (Baier and Bergstrand, 2004)
- Customs unions should have the largest effects (Larch and Yotov, 2024, p. 1787)
- Borders are large frictions in quantities (McCallum, 1995; Anderson and van Wincoop, 2003) and prices (Engel and Rogers, 1996), partly country heterogeneity (Gorodnichenko and Tesar, 2009)

## Geography takes a growing role

- Trade costs and market access are modelled from geography (Krugman, 1991; Donaldson, 2015; Redding and Rossi-Hansberg, 2017)
- In Germany, geography set which states joined (Keller and Shiue, 2014) and their bargaining power through routes (Huning and Wolf, 2026)
- Old borders persisted in trade long after unification (Wolf, 2009)

## What this setting adds

- Several agreements forming *at once* in one space, not one at a time
- Three designs: two full customs unions and one standstill pact
- City-level prices, so borders are measured at any distance
- Five years (1828–33) before a full merger: the interaction of blocs is visible before it is resolved
- Building or stumbling blocks? (Bhagwati, 1993; Freund and Ornelas, 2010)
  - Ex post, building blocks: the blocs merged within five years, outsiders followed (Baldwin, 1993), political union came in 1871
  - What did they do to markets in the meantime?

# Historical background

# Before 1815: about 1,800 customs lines, then Napoleonic consolidation

- Late 18th century: over 300 states (Henderson, 1939, p. 1) and about 1,800 customs lines; tolls on rivers, roads and at town gates (Hahn, 1984, p. 10)
  - Customs borders were a matter for the Emperor and Electors, not the states (Price, 1949)
- 1806: the Empire ends; Napoleon reorganises Germany, the southern states enlarged and compact (Hahn, 1984, p. 11; Henderson, 1939, p. 2)
- Bavaria (1807), Württemberg (1808), Baden (1812): internal duties abolished, one border tariff, revenue as the aim (Hahn, 1984, pp. 12–13; Henderson, 1939, p. 15)
  - Many states could not enforce their customs laws, given the size and shape of their territory (Price, 1949)
  - The new borders also cut old regional trade ties (Hahn, 1984, p. 13)
- Institutions mattered for prices: gaps fell by about 6 points (31%) where a city joined a larger state *and* the state reformed its tariffs (Albers and Pfister, 2023, pp. 404, 412); French-rule reforms (1792–1815) cut later gaps by about a quarter (Keller and Shiue, 2020)

# 1815: Vienna draws the borders for politics, not trade

- Economic questions were secondary at the Congress (Price, 1949); the German Confederation had no competence over trade (Hahn, 1984, p. 15); Art. 19 promised free commerce, the Diet never acted (Treitschke, 1906, pp. 615–616)
- Britain's intervention moved Prussia to the Rhine: two halves, east and west, separated by Hanover and the Hessian states (Huning and Wolf, 2026)
- Prussia now controlled the main waterways (Rhine, Elbe) and the transit routes of its neighbours (Huning and Wolf, 2026)
- Bavaria too was split: the Palatinate west of the Rhine (Henderson, 1939, p. 7; Kunz, 2008)




States in 1820, main rivers, the 75 grain cities.

Source: author's creation, based on IPUMS MOSAIC data provided by MPIDR

# 1818: Prussia builds one customs territory, for revenue

- Customs law of 26 May 1818: internal duties abolished, one external tariff (Hahn, 1984, p. 20)
- The aim was revenue: colonial goods gave about 70% of import revenue in 1819 (Hahn, 1984, p. 22); transit through Prussia was taxed
- 1819–1830: the enclaves inside Prussia join its system by treaty (Henderson, 1934, p. 3)
- The Prussian wall forced its neighbours to rethink (Hahn, 1982), but small states gained little from their own borders: guarding them ate the revenue (Hahn, 1984, p. 35; Dumke, 1984, p. 86)
  - Nassau's low-tariff system (1822) paid, but was hard to run (Hahn, 1982, pp. 67–68)
  - The Hessian systems of 1824 failed (Hahn, 1984, p. 38)
  - Smuggling into Prussia from the small states' bonded warehouses was rife (Treitschke, 1906, pp. 610–611)

# 1828: three unions in one year

| Union | Treaty | Members (in force) |
|---|---|---|
| SGCU (South German) | 18 Jan 1828 | Bavaria (Palatinate from 1830), Württemberg, the Hohenzollerns; from 1 Jul 1828 |
| PHC (Prussian–Hessian) | 14 Feb 1828 | Prussia, Hesse-Darmstadt, the Anhalt duchies, Prussian enclaves; from 1 Jul 1828 |
| CGCU (Central German) | 24 Sep 1828 | Saxony, Hanover, Hesse-Kassel, Brunswick, Oldenburg, Bremen, Frankfurt, the Thuringian and Reuß states; Nassau from 1829 |

- Hesse-Darmstadt brought Prussia no revenue; its value was the pressure on the others (Ploeckl, 2015, p. 288)
- The CGCU was the reply: a defensive pact, backed by Austria, between Prussia's two halves (Henderson, 1934, p. 5)
- Outside all three: Baden, the Mecklenburgs, Hamburg and Lübeck (Helmreich, 1949, p. 153)
- Treaties: Martens, *Nouveau Recueil* 7.2, pp. 529, 550; *British and Foreign State Papers* 15, p. 1266; dates in force: Kunz (2008)

# Each treaty touched a different part of the cost

| | PHC and SGCU | CGCU |
|---|---|---|
| Internal duties | removed | kept; grain duty-free between members from 1 Jan 1829, if it did not cross a foreign state (Art. XIV) |
| External tariff | common (Prussian, or Bavarian) | none; each state kept its own |
| Transit duties | common | kept low; a road programme |
| Commitment | shared revenue; sovereign partners | members may not join another union |

- The CGCU was "neither a customs union nor a free trade area" (Ploeckl, 2015, p. 289)
- Table: Hahn (1984, pp. 41, 46); Helmreich (1949, p. 153); CGCU treaty, Art. XIV (*British and Foreign State Papers* 15)

# Why states joined: revenue and geography more than markets

- Small states joined to fill empty treasuries and to escape a bad position behind customs borders (Dumke, 1984, p. 92)
- The main gain was cheaper revenue collection; expected economic gains were not a sufficient reason (Dumke, 1984, pp. 92–95)
- Border cost rises with length, revenue with area: the border-to-area ratio sets the cost of collection (Kühne, 1836; Dumke, 1984, pp. 85–89)
- Prussia negotiated state by state, using the externalities on those outside (Ploeckl, 2015, pp. 278, 297)
- Location decided timing: states whose trade passed through Prussia joined early; states with seaports joined last (Keller and Shiue, 2014; Huning and Wolf, 2026)
  - Today too, transit structure, not landlockedness alone, sets trade costs (Endale, 2026)
- For this paper: membership is not random; it follows geography and fiscal need

**Border-to-area ratio and cost of collection**

| | B/A | Cost, % of revenue |
|---|---|---|
| Zollverein | 0.13 | 15 |
| Prussia | 0.21 | 15–20 |
| Bavaria | 0.24 | 25 |
| Württemberg | 0.47 | 43 |
| Palatinate | 0.48 | 150 |
| Hesse-Kassel | 0.77 | ≈100 |
| Hesse-Darmstadt | 0.91 | ≈100 |

(Dumke, 1984, 87, Table 2)

Zollverein 1836: external border 1,064 miles, barely longer than Prussia's 1,073 in 1819; 782 miles of internal border gone (Kühne, 1836)

# 1829–1834: the blocs merge into the Zollverein

- 27 May 1829: commercial treaty between the PHC and the SGCU, a first step to a merger (Huning and Wolf, 2026)
- 1830–31: the Belgian revolution lowers Rhine tariffs; overland transit through Hesse-Kassel collapses (Huning and Wolf, 2026)
- August 1831: Hesse-Kassel joins the PHC (in force 1832), splitting the CGCU in two (Ploeckl, 2015, p. 291)
- 1833 treaties, in force 1 January 1834: the Zollverein (Jopp and Spoerer (2024, p. 56))
- Late joiners (accession treaties): Baden 1835, Brunswick 1841, Hanover 1851, Oldenburg 1852; Hamburg and Bremen only after 1871 (Huning and Wolf, 2026)

# Market integration: the logic

# A price gap bounds the cost of moving grain between two markets

$$|\log p_{it} - \log p_{jt}| \leq \underbrace{\tau_{ij}^{\text{transport}}}_{\text{distance, terrain}} + \underbrace{\tau_{ijt}^{\text{duty}}}_{\text{customs}} + \underbrace{\tau_{ijt}^{\text{other}}}_{\text{currency, information}}$$

- Law of one price with trade costs: arbitrage keeps the gap inside a band
- Market integration = a narrower band, or prices that move together within it (Federico et al., 2021, p. 6)
- The unions act on $\tau^{\text{duty}}$; geography on $\tau^{\text{transport}}$; 34 states and 4 free cities still minted their own coins in 1834 (Reckendrees, 2023, p. 6)

# What a price gap can and cannot tell us

## Can

- Bound the cost of trade between two markets
- Measure integration where trade flows are not recorded (1820s Germany)
- Follow integration year by year, for many pairs

## Cannot

- Show whether trade took place: a gap measures trade costs "only if the two markets trade" (Federico, 2012, p. 476)
- Separate a fall in costs from a fall in shocks, without controls
- Tell which direction grain moved
- Stand for all goods: grain is a thick, homogeneous market, but bulky and often regulated (Federico, 2012, p. 486)

- Trade flows carry information and make arbitrage possible; once a link is established, its prices follow market rules

# Grain in the 1820s: bulky, mostly regional, carried by river and road

- River freight cost 0.6–1.5% of road freight per ton-km; paved roads cut road costs to about a quarter (Huning and Wolf, 2026)
- Most European trade moved overland or on inland waterways, not by sea (Chilosi et al., 2013, p. 47); grain was a main cargo of roads and canals (Brakensiek, 2024, pp. 461–464)
- Rhine tolls and staple rights pushed north–south trade to the Elbe and the road from Magdeburg to Nuremberg (Albers and Pfister, 2023, p. 408)
- Rheinhessen exported about 36% of its grain; Kurhessen was far from self-sufficient; Nassau sold to the Prussian Rhineland (Hahn, 1982, pp. 36–41)

# Prussia's grain duty was per volume: 7% of the price of wheat, 21% of oats

- 1818 tariff, per Scheffel: wheat 1 7/8 Sgr, rye and barley 5/8, oats 5/16 (Oechelhäuser, 1851, p. 36)
- From 1825 every grain paid 5 Sgr per Scheffel; the Zollverein kept the rate (Oechelhäuser, 1851, p. 36)
- Tolls were per weight too: in the SGCU road and river tolls were capped at 2 Pfennig per Centner and hour (treaty of 18 Jan 1828, Art. XXVIII)
- A specific duty is a larger share of the price of cheap grains: if duties bind, union contrasts should be ordered oats > barley > rye > wheat

**The 1825 duty against grain prices, 1828–33**

| Grain | Median price (g Ag/L) | Duty, % of price | Log points |
|---|---|---|---|
| wheat | 0.68 | 7 | 0.07 |
| rye | 0.48 | 10 | 0.10 |
| barley | 0.35 | 14 | 0.13 |
| oats | 0.24 | 21 | 0.19 |

Median price across the 75 cities; 1 Scheffel = 55 litres.

Source: author's calculation from Oechelhäuser (1851, p. 36) and the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007)

# What we know for Germany

# Integration came early and in steps

| Study | Finding |
|---|---|
| Fremdling and Hohorst (1979, p. 79) | an integrated Prussian rye market probably existed well before 1821; little further integration to 1865 |
| Albers and Pfister (2023, pp. 404, 417) | the 1818 Prussian law cut trade costs by about 43%; pairs in old territories drifted apart in 1819–30 |
| Uebele (2010, p. 2) | North German cities outside the Zollverein moved closely together |
| Uebele and Gallardo-Albarrán (2015, p. 71) | distance effects fell by about 22% (rye) and 16% (wheat) from the 1820s to the 1840s |
| Federico (2012, p. 491) | massive price convergence in Europe from Waterloo to about 1870 |

# The Zollverein mattered, but slowly and alongside transport

| Study | Finding |
|---|---|
| Shiue (2005, p. 158) | the Zollverein border effect was similar for early and late joiners and grew over time |
| Keller and Shiue (2008, p. 29) | on price gaps: railways about $-0.22$, the customs union about $-0.13$; currency agreements also substantial |
| Keller and Shiue (2014, p. 3) | ignoring that accession is endogenous biases the Zollverein's effect downward |
| Keller and Shiue (2020) | French-imposed institutions lowered city-to-city price gaps |
| Wolf (2009, pp. 846, 877) | even the Empire before 1914 was poorly integrated |

## Where this paper sits

- Integration *within* states (reforms, 1800–1830) and the Zollverein *between* states (1834 on) are well studied (Albers and Pfister, 2023, p. 404)
- The five years in between, when three blocs competed, are rarely studied as such
- City-level prices and year-by-year territory and customs status make the blocs' borders visible

# Predictions

| Pair | What the treaty did | Predicted gap |
|---|---|---|
| Within PHC, within SGCU | internal duties removed, common tariff | falls |
| Within CGCU | no common tariff; grain freed between members from 1829 | falls a little |
| Union and non-member | external tariff applies | rises relative to within |
| Two rival unions | two different tariff walls; PHC–SGCU grain free from 1 Jan 1830 | rises more than a border; PHC–SGCU falls from 1830 |

- Preferential liberalisation closes the price gap with partners and widens it with outsiders (Baldwin and Wyplosz, 2015, ch. 5)
- Customs unions should have the largest effects (Larch and Yotov, 2024, p. 1787)

# Measurement and setting

# 75 cities, four grains, prices in grams of silver per litre

| | Pairs | Cities | Same state | Obs. | Price gap | Return gap |
|---|---|---|---|---|---|---|
| *Panel A: customs relationship, 1828–1833* | | | | | | |
| Within a union: CGCU-CGCU | 75 | 17 | 0.41 | 1,250 | 0.146 | 0.123 |
| Within a union: PHC-PHC | 142 | 29 | 0.82 | 2,914 | 0.126 | 0.102 |
| Within a union: SGCU-SGCU | 76 | 17 | 0.95 | 742 | 0.098 | 0.073 |
| Between two unions: CGCU-PHC | 178 | 39 | 0.00 | 3,248 | 0.154 | 0.114 |
| Between two unions: CGCU-SGCU | 16 | 12 | 0.00 | 146 | 0.148 | 0.143 |
| Between two unions: PHC-SGCU | 25 | 16 | 0.00 | 280 | 0.160 | 0.100 |
| Union and non-member: CGCU-None | 60 | 21 | 0.00 | 1,258 | 0.174 | 0.126 |
| Union and non-member: None-PHC | 61 | 34 | 0.00 | 1,266 | 0.175 | 0.104 |
| Union and non-member: None-SGCU | 11 | 8 | 0.27 | 114 | 0.132 | 0.140 |
| Neither in a union: None-None | 31 | 12 | 0.48 | 578 | 0.203 | 0.077 |
| *Panel B: by grain, 1818–1833* | | | | | | |
| Wheat | 595 | 69 | 0.40 | 8,696 | 0.152 | 0.096 |
| Rye | 596 | 66 | 0.33 | 8,678 | 0.152 | 0.121 |
| Barley | 474 | 59 | 0.39 | 6,825 | 0.169 | 0.127 |
| Oats | 447 | 57 | 0.37 | 6,670 | 0.186 | 0.138 |
| Full sample, 1818–1833 | 682 | 75 | 0.37 | 30,869 | 0.163 | 0.119 |

- After 1828 gaps were smallest within a union (0.10–0.15) and largest where neither city was in one (0.20); this ordering predates the unions

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007); customs status: author's coding from the treaties.

# Making prices comparable: grams of silver per litre

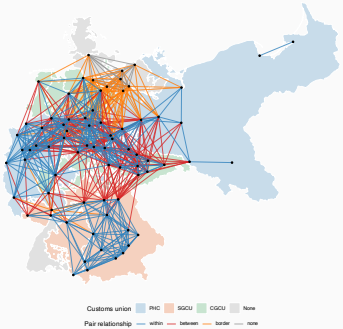
## Sources

- 70 cities: Albers and Pfister (2024), as published, with their own measure and currency conversions
- Arnsberg, Chemnitz, Stade: Jacks (2004, *J. European Economic History*); Jacks (2005); from Kopsidis (1995), Strauss (1963), Oberschelp (1986)
- Nine Bavarian cities: Shiue and Keller (2007), from Seuffert (1857)
- One source per city and grain: the longest series, Albers and Pfister on a tie

## Conversion

- Many currencies: even in 1834 the German states minted on several standards (12-, 13 1/3- and 14-Taler, 24-Gulden), with different silver per coin (Sprenger, 1991, p. 160)
- Many measures: the Scheffel, Himten or Malter differed from state to state and town to town (Krüger 1830, *Vollständiges Handbuch der Münzen, Maße und Gewichte aller Länder der Erde*)
- Every price is converted to grams of fine silver per litre: price $\times$ silver per coin $\div$ litres per measure

# Every city pair has a customs relationship

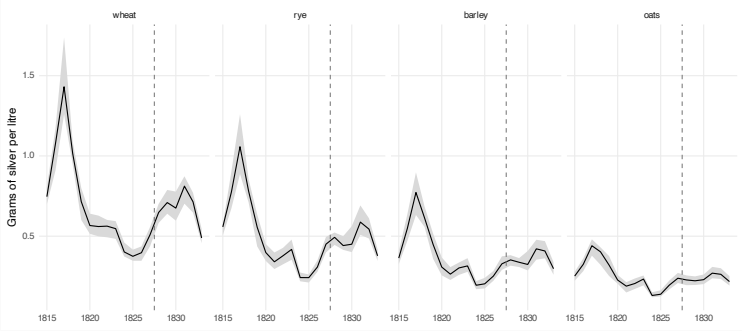



- Within 200 km most pairs cross a customs border; CGCU–PHC is the largest group (178 pairs), pairs outside every union are few and northern

Source: author's creation, based on IPUMS MOSAIC data provided by MPIDR; customs status: author's coding from the treaties.

# The macro picture

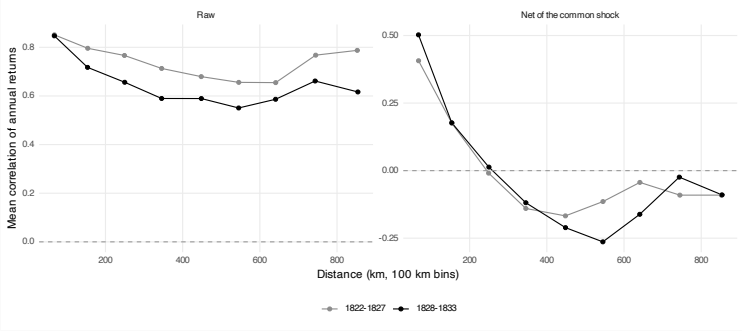
# Aggregate shocks shrank after 1828 and prices became more local




- The Germany-wide shock was about 30% smaller after 1828 (SD of the common annual return 0.25 $\rightarrow$ 0.18); its share of price variance fell from 0.51 to 0.34

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007).

# Net of the common shock, nearby markets moved together more after 1828

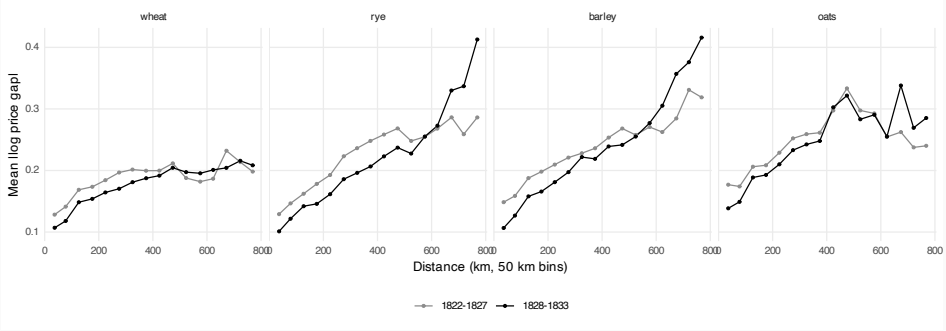



- Net of the common shock (each year's mean price change), cities within 100 km moved together more (0.41 $\rightarrow$ 0.50) and distant regions moved apart

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007).

# The spatial picture

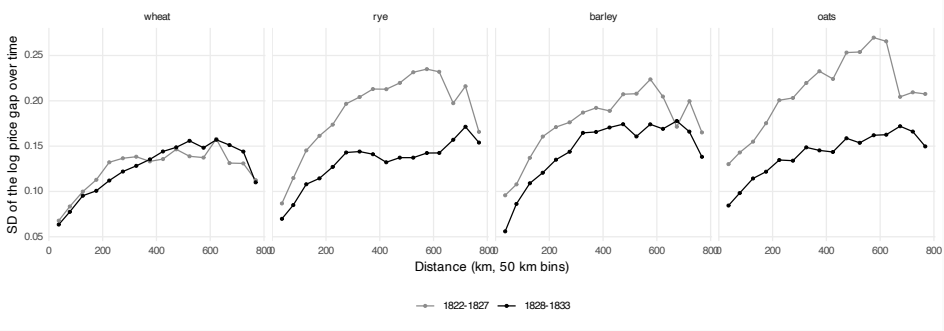
# Distance mattered more after 1828: near pairs converged, distant ones did not




- Gaps fell for near pairs, not distant ones: the slope of the gap on log distance rose from 0.050 to 0.069, in every grain

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007).

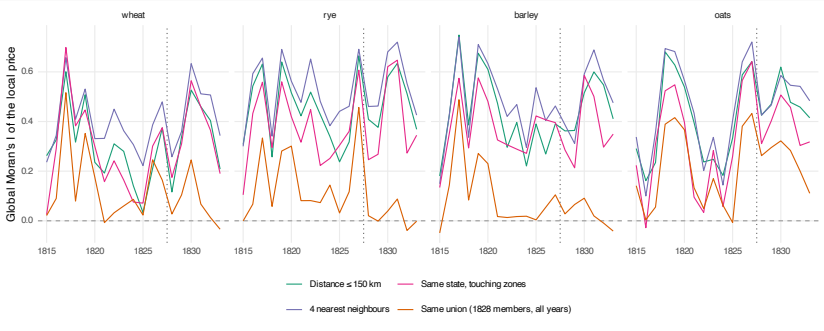
# Price gaps became steadier at every distance (not for wheat)




- Gaps fluctuated less from year to year after 1828 for rye, barley and oats, at every distance; wheat, traded more internationally, did not

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007).

# Local prices clustered by geography, not by union




- Within 150 km, cities of the same state shared price levels more after 1828 (0.36 $\rightarrow$ 0.46); cities of the same union but different states did not (0.08 $\rightarrow$ 0.07)

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007); customs status: author's coding from the treaties.

# Unions and states

# Every kind of pair converged, and the ordering predates the unions




- Every group converged by 2–4 points; the ordering within $<$ between $<$ border $<$ neither was there before 1828

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007); customs status: author's coding from the treaties.

# In raw means the union adds at most a point; the state is a stable level

| Mean price gap, pairs $\leq$ 200 km | 1822–27 | 1828–33 | change |
|---|---|---|---|
| Within a union | 0.153 | 0.127 | −0.026 |
| Between two unions | 0.172 | 0.156 | −0.016 |
| Union and non-member | 0.205 | 0.173 | −0.032 |
| Neither in a union | 0.240 | 0.202 | −0.038 |
| Same state | 0.156 | 0.131 | −0.025 |
| Same union, different states | 0.182 | 0.145 | −0.037 |
| Different unions or non-member, different states | 0.189 | 0.164 | −0.024 |

- Within-union gaps fell about one point more than between-union gaps; same-state pairs stayed about three points below the rest throughout

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007); customs status: author's coding from the treaties.

# The raw union contrasts sit in barley and oats

| Change in mean gap, 1822–27 → 1828–33 | wheat | rye | barley | oats |
|---|---|---|---|---|
| Within a union | −0.015 | −0.021 | −0.040 | −0.028 |
| Between two unions | −0.025 | −0.021 | −0.013 | −0.004 |
| Same union, different states | −0.025 | −0.013 | −0.050 | −0.059 |
| Other cross-state pairs | −0.025 | −0.030 | −0.026 | −0.016 |
| Size of the common shock | +0.062 | −0.104 | −0.098 | −0.159 |

- The extra fall within unions comes from barley and oats, the grains whose common shock also shrank most

Source: author's calculations from the price data of Albers and Pfister (2024), Jacks (2005), Shiue and Keller (2007); customs status: author's coding from the treaties.

# Results

# Specification

$$\underbrace{|\log p_{it}^{g} - \log p_{jt}^{g}|}_{\text{price gap}} = \beta' \mathbf{D}_{ijt} + \alpha_{ij} + \gamma_{t}^{g} + \varepsilon_{ijt}^{g}$$

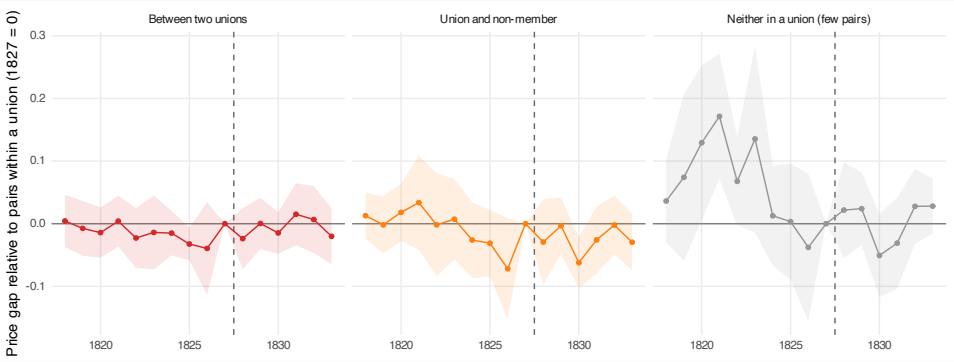
- $\mathbf{D}_{ijt}$: the pair's customs relationship in year $t$: within a union, across a union boundary (two unions, or a union and a non-member); all zero before 1828
- $\alpha_{ij}$: pair effects absorb distance, terrain and the two states
- $\gamma_{t}^{g}$: year $\times$ grain effects absorb the common shock; second specification: city $\times$ grain $\times$ year effects for both cities
- Pairs in different states at most 200 km apart, 1818–1833 (429 pairs, 65 cities); reference: neither city in a union; dyad-robust standard errors

# Within a union and across a union boundary, gaps changed alike

| | Change in gap relative to pairs outside every union |
|---|---|
| Within a union | 0.046* (0.026) |
| Across a union boundary | 0.043* (0.025) |
| Across − within | -0.003 (0.012) |
| Between two unions − within | 0.008 (0.012) |

- Both groups' gaps fell about 4–5 points *less* than for pairs outside every union
- The difference between them is zero: −0.003 (s.e. 0.012)

# No break at 1828: rival-union pairs track within-union pairs throughout



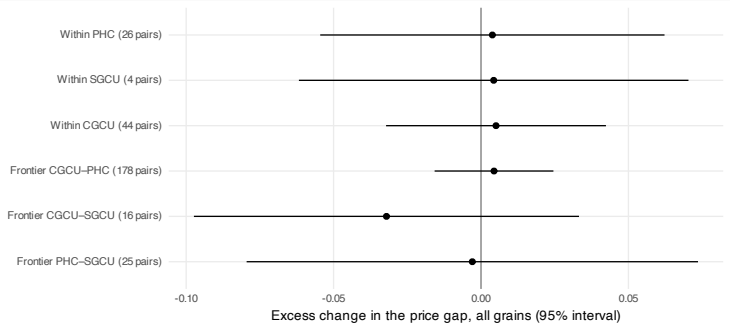
# By union: no integration inside, no widening at the frontiers

Did a union's members converge more with each other than with outsiders? Did a frontier widen more than the two unions' ties with outsiders? (Pairs in different states, at most 200 km apart.)

| Pair | Pairs | Predicted | Relative to ties with outsiders | Net of city shocks |
|---|---|---|---|---|
| Within PHC | 26 | $< 0$ | -0.006 (0.025) | 0.004 (0.030) |
| Within SGCU | 4 | $< 0$ | 0.108*** (0.011) | 0.004 (0.034) |
| Within CGCU | 44 | $\leq 0$ | 0.027* (0.015) | 0.005 (0.019) |
| Frontier CGCU–PHC | 178 | $> 0$ | 0.026*** (0.010) | 0.004 (0.010) |
| Frontier CGCU–SGCU | 16 | $> 0$ | 0.036* (0.020) | -0.032 (0.033) |
| Frontier PHC–SGCU | 25 | $> 0$ | 0.068*** (0.012) | -0.003 (0.039) |

- Inside: no union's members converged more with each other than with outsiders (SGCU: 4 pairs)
- Frontiers look 2.5–7 points wider, but this coincides with the unions' ties with outsiders converging (previous slide); PHC–SGCU grain was duty-free from 1830. Net of city shocks: all zero

# Net of each city's own shocks, no pair-specific union difference remains




- City × grain × year effects: what is left is the pair's *excess* over the borders of its two cities
- CGCU–PHC frontier: 0.004 (0.010); the 95% bound, 0.024, is about the gap associated with doubling the distance between two cities (0.025)

# Why naive comparisons show union differences: the comparison group

| Grain | Pairs, 1818–27 | Pairs, 1828–33 | Obs., 1818–27 | Obs., 1828–33 |
|---|---|---|---|---|
| wheat | 355 | 8 | 3214 | 44 |
| rye | 400 | 15 | 3642 | 86 |
| barley | 288 | 8 | 2537 | 44 |
| oats | 282 | 8 | 2519 | 44 |

- After 1828 the pairs outside every union are Hamburg, Rendsburg, the Mecklenburg towns and Detmold
- Coastal states stayed out because their ports made Prussian tariffs matter least (Huning and Wolf, 2026); their prices moved together on their own (Uebele, 2010, p. 2)
- A border effect measured against a group that differs anyway (Gorodnichenko and Tesar, 2009)

# Interpretation

# What the evidence says

**Grain markets integrated after 1828 along geographic lines; union membership is associated with no detectable additional integration**

- Smaller common shocks, closer co-movement of neighbours, a steeper distance gradient
- Within a union and across its boundary, gaps changed alike
- No detectable union difference for pairs of markets, inside the unions or at the rival frontiers
  - CGCU–PHC frontier: at most about the gap associated with doubling the distance
  - Prussia–Hesse-Darmstadt: the 1828 treaty removed a duty of 7–21% of the price; their gaps changed no differently from those with outsiders (within about $\pm 5$ points)
- Comparisons with non-members mislead: after 1828 they are a few coastal markets that stayed out (Huning and Wolf, 2026)
- By grain, the raw union contrasts and the one frontier excess net of city shocks (oats, A1) sit in the cheap grains, in line with a per-volume duty

# Why might the unions show no price differences?

1. **The unions formed along an existing price map**
   - The ranking of gaps (within $<$ between $<$ border $<$ neither) predates 1828, and local prices cluster by state rather than by union
   - Membership follows geography and fiscal need (Dumke, 1984; Keller and Shiue, 2014; Huning and Wolf, 2026)
2. **Transport, not duties, may have been the binding cost**
   - Gaps rise with distance, more steeply after 1828; river freight cost a fraction of road freight (Huning and Wolf, 2026)
3. **Merchants may have expected the blocs to merge**
   - PHC–SGCU treaty in May 1829, grain duty-free between the two from 1830, merger agreed in 1833
4. **Borders were porous**
   - Smuggling from bonded warehouses (Treitschke, 1906, pp. 610–611); small states could not enforce their tariffs (Price, 1949)

# Implications

- Ex post, the blocs appear as building blocks: they merged by 1834, outsiders followed, and political union came in 1871. Over 1828–33, membership is associated with little change in grain price gaps
- When membership is selected, the comparison group shapes the estimated difference; here, the non-members are a handful of coastal markets
- A tariff on paper need not be the binding cost: removing a duty of 7–21% of the price between Prussia and Hesse-Darmstadt coincides with no detectable extra convergence

## Summary

- Three competing trade blocs in one economic space, 1828–1833
- Markets integrated, but regionally and by distance
- Membership in an agreement is not associated with extra integration of a pair of markets
- No detectable bloc difference for pairs of markets: at most about the gap associated with doubling the distance; no extra convergence where a 7–21% duty was removed
- Comparisons with non-members mislead when non-membership is selected (here: the coast)
- Next for this paper: route-based distance (rivers, roads), the Bavarian tariff and the 1825 duty rise

# Work in progress

1. **Exclaves and their implications**
   - Exclaves joined their neighbours' customs systems by separate treaties (e.g. Meisenheim 1829, Lichtenberg and Birkenfeld 1830): borders that moved without moving the state
2. **Monetary fragmentation under monetary standards**
   - Several coin standards side by side (12-, 13 1/3-, 14-Taler, 24-Gulden): exchange rates, silver content and mints as a second layer of trade costs
3. **More price data**
   - More cities and higher-frequency (monthly, weekly) series
4. **Population statistics**
   - The Zollverein censuses from 1834 onward: market size and demand alongside prices

# Thank you

# Appendix

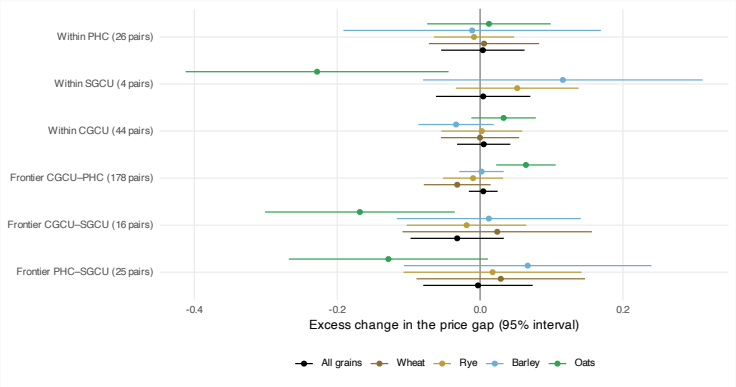
A1 **By grain, net of city shocks:** the excess of each union and frontier, grain by grain

A2 **All specifications:** by union and frontier, with pair and year $\times$ grain effects (R1), net of city shocks (R2), without rye, and by grain

A3 **Excesses compared:** within and frontier excesses computed from R1 and estimated directly in R2

A4 **Data and coding:** prices, territory, customs membership

# A1. By grain, net of city shocks



# A2. All specifications

| Model: | R1 all grains (1) | R2 city-year FE (2) | R1 without rye (3) | Wheat (4) | Rye (5) | Barley (6) | Oats (7) |
|---|---|---|---|---|---|---|---|
| *Variables* | | | | | | | |
| Within PHC | 0.0312 | 0.0039 | -0.0363** | 0.0299 | 0.1302*** | -0.0771* | -0.0637*** |
| | (0.0279) | (0.0298) | (0.0176) | (0.0343) | (0.0342) | (0.0416) | (0.0110) |
| Within SGCU | 0.0626** | 0.0043 | 0.0095 | | 0.1465*** | -0.0068 | -0.0019 |
| | (0.0270) | (0.0337) | (0.0190) | | (0.0316) | (0.0187) | (0.0206) |
| Within CGCU | 0.0517* | 0.0051 | -0.0063 | 0.0321 | 0.1373*** | -0.0130 | -0.0469* |
| | (0.0288) | (0.0190) | (0.0199) | (0.0334) | (0.0324) | (0.0441) | (0.0281) |
| PHC border | 0.0373 | | -0.0086 | 0.0314 | 0.1086*** | -0.0295** | -0.0290** |
| | (0.0233) | | (0.0131) | (0.0251) | (0.0287) | (0.0133) | (0.0136) |
| SGCU border | -0.0453* | | -0.0831*** | -0.1302*** | 0.0060 | -0.1395*** | 0.0322** |
| | (0.0241) | | (0.0159) | (0.0335) | (0.0203) | (0.0162) | (0.0141) |
| CGCU border | 0.0250 | | -0.0153 | 0.0261 | 0.0814*** | -0.0267 | -0.0474*** |
| | (0.0216) | | (0.0115) | (0.0271) | (0.0227) | (0.0179) | (0.0112) |
| Frontier CGCU–PHC | 0.0572** | 0.0044 | 0.0096 | 0.0309 | 0.1324*** | -0.0030 | -0.0010 |
| | (0.0254) | (0.0103) | (0.0153) | (0.0349) | (0.0307) | (0.0221) | (0.0128) |
| Frontier CGCU–SGCU | 0.0256 | -0.0321 | -0.0141 | 0.0240 | 0.0766** | -0.0094 | -0.0811*** |
| | (0.0299) | (0.0333) | (0.0232) | (0.0454) | (0.0326) | (0.0216) | (0.0242) |
| Frontier PHC–SGCU | 0.0644** | -0.0029 | 0.0180 | 0.0740** | 0.1391*** | -0.0032 | -0.0365** |
| | (0.0258) | (0.0391) | (0.0164) | (0.0336) | (0.0365) | (0.0261) | (0.0160) |
| *Fixed-effects* | | | | | | | |
| Pair | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year × grain | Yes | | Yes | | | | |
| City $i$ × grain × year | | Yes | | | | | |
| City $j$ × grain × year | | Yes | | | | | |
| Year | | | | Yes | Yes | Yes | Yes |
| *Fit statistics* | | | | | | | |
| Observations | 19,294 | 38,332 | 13,434 | 5,156 | 5,860 | 4,157 | 4,121 |
| $R^2$ | 0.29002 | 0.70264 | 0.31138 | 0.52563 | 0.36382 | 0.39067 | 0.49357 |

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Pairs in different states at most 200 km apart; 1818–1833; spelt excluded. Reference: neither city in a union. Column 2: pairs entered in both orders, city × grain × year effects for both cities; within and frontier coefficients are excesses over the borders (see text); standard errors two-way clustered by the two cities.

# A3. Within and frontier excesses, R1 and R2

| | R1 (from the nine coefficients) | R2 (estimated directly) |
|---|---|---|
| Frontier excess CGCU–PHC | -0.005 (0.023) | 0.004 (0.010) |
| Frontier excess CGCU–SGCU | 0.046 (0.030) | -0.032 (0.033) |
| Frontier excess PHC–SGCU | 0.072*** (0.027) | -0.003 (0.039) |
| Within excess PHC | -0.043 (0.039) | 0.004 (0.030) |
| Within excess SGCU | 0.153*** (0.026) | 0.004 (0.034) |
| Within excess CGCU | 0.002 (0.024) | 0.005 (0.019) |

# A4. Data and coding

- Prices: sources and conversion on slide 21
- Territory: MOSAIC historical GIS (IPUMS, MPIDR), contiguous zones by state and year; each city placed in its zone
- Customs membership: coded from the treaty texts and Kunz (2008), by the year the membership took effect
  - CGCU coded from 1828, although its grain clause (Art. XIV) applies from 1 Jan 1829
  - Nassau in the CGCU from 1829; the Palatinate in the SGCU from 1830; Hesse-Kassel in the PHC from 1832
- Regression sample: pairs in different states, at most 200 km apart, 1818–1833, spelt excluded

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